

GROSSETO, Italy

WMO: 162060

Lat: 42.760N

Lon: 11.072E

Elev: 5

StdP: 101.27

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.8	-0.2	-10.4	1.5	5.9	-7.8	1.9	5.7	15.0	9.2	13.2	8.4	3.7	40	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.4	34.4	21.5	32.9	21.4	31.6	21.2	25.7	29.2	24.7	28.5	23.8	27.9	5.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	19.7	27.9	23.6	18.5	27.0	22.6	17.3	26.3	79.1	29.2	75.0	28.6	71.3	28.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.8	10.7	9.2	-4.8	37.1	1.9	1.5	-6.2	38.2	-7.3	39.1	-8.4	39.9	-9.8	41.0
			-5.5	27.3	1.8	1.3	-6.8	28.3	-7.8	29.0	-8.9	29.7	-10.2	30.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	7.9	8.3	10.7	13.5	17.7	22.0	24.8	25.1	21.2	17.4	12.6	8.8
	DBStd	6.69	2.89	2.95	2.53	2.48	2.40	2.71	2.16	2.23	2.58	2.73	3.28	3.16
	HDD10.0	236	77	60	22	3	0	0	0	0	0	1	14	59
	HDD18.3	1554	323	280	236	146	41	3	0	0	6	50	174	296
	CDD10.0	2379	12	13	44	108	238	361	458	335	228	91	22	
	CDD18.3	656	0	0	0	1	21	114	200	209	91	20	1	0
	CDH23.3	6674	0	0	1	12	224	1183	2259	2222	688	84	1	0
	CDH26.7	2633	0	0	0	1	40	421	1007	997	164	4	0	0
Wind	WSAvg	3.8	3.9	4.1	4.2	3.8	3.4	3.5	3.7	3.5	3.8	3.7	4.0	4.0
Precipitation	PrecAvg	653	58	51	54	49	39	28	16	34	72	82	102	67
	PrecMax	917	120	124	159	125	80	81	80	144	269	209	356	175
	PrecMin	423	1	0	2	1	0	1	0	0	0	0	11	0
	PrecStd	136	40	31	38	29	21	22	17	37	58	53	82	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.9	17.8	21.3	24.9	30.0	35.2	35.8	37.0	32.0	27.0	21.7	17.9
		MCWB	13.3	11.7	13.4	15.6	18.7	21.2	21.5	21.6	19.9	18.8	17.2	14.6
	2%	DB	15.2	16.1	19.0	22.8	27.6	32.4	34.1	34.9	30.0	25.3	20.1	16.2
		MCWB	12.3	11.4	12.5	14.9	17.8	20.3	21.2	21.7	20.6	19.3	16.6	13.4
	5%	DB	14.2	15.1	17.8	21.1	25.8	30.8	32.9	33.0	28.6	24.1	19.1	15.2
		MCWB	11.5	11.3	12.4	14.6	17.5	20.1	21.3	21.9	20.1	19.0	15.9	12.5
	10%	DB	13.2	14.0	16.3	19.5	24.1	29.1	31.6	31.4	27.1	22.8	17.9	14.1
		MCWB	10.6	10.8	11.9	14.1	17.1	20.0	21.2	21.8	19.9	18.5	15.1	11.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.4	14.0	15.7	18.2	21.9	25.3	27.0	27.7	25.5	22.5	19.1	15.5
		MCDB	15.8	15.7	18.3	21.9	25.6	29.4	30.4	30.9	27.9	24.4	21.0	17.0
	2%	WB	13.2	13.0	14.6	16.8	20.4	23.7	25.6	25.9	24.1	21.3	17.5	14.2
		MCDB	14.7	14.9	17.1	20.3	24.2	28.0	29.4	29.3	26.6	23.6	19.3	15.8
	5%	WB	12.3	12.2	13.7	15.8	19.3	22.6	24.6	24.9	22.8	20.3	16.5	13.2
		MCDB	13.8	14.2	16.2	19.2	23.2	27.2	28.7	28.6	26.0	22.8	18.4	14.7
	10%	WB	11.1	11.3	12.8	14.9	18.2	21.6	23.6	23.9	21.6	19.3	15.3	12.1
		MCDB	12.9	13.4	15.4	18.2	22.4	26.3	28.1	28.0	25.1	21.8	17.4	13.9
Mean Daily Temperature Range	5% DB	MDBR	9.0	10.0	10.7	11.2	11.7	12.5	12.7	12.4	11.2	10.1	8.6	8.7
		MCDBR	10.0	11.4	13.1	13.6	14.3	14.7	14.4	14.5	12.6	11.3	9.0	9.4
		MCWBR	7.7	8.3	8.4	8.0	7.2	6.3	6.0	6.2	6.5	7.1	6.5	7.2
	5% WB	MCDBR	8.6	9.7	10.6	11.4	12.4	12.3	12.0	11.9	10.2	9.4	7.8	8.3
		MCWBR	7.5	7.9	8.2	8.1	8.2	7.2	7.3	7.3	7.2	7.2	6.4	7.2
Clear-Sky Solar Irradiance	taub	0.339	0.358	0.392	0.425	0.431	0.429	0.421	0.419	0.406	0.393	0.369	0.338	
	taud	2.458	2.387	2.298	2.223	2.251	2.302	2.326	2.348	2.370	2.389	2.392	2.467	
	Ebn at Noon	794	835	845	843	846	847	849	840	822	783	743	758	
	Edh at Noon	76	95	116	135	135	128	124	118	107	94	81	70	
All-Sky Solar Radiation	RadAvg	1.69	2.60	3.83	5.14	6.33	7.20	7.36	6.39	4.72	3.08	1.86	1.49	
	RadStd	0.17	0.26	0.45	0.37	0.50	0.35	0.33	0.32	0.24	0.27	0.17	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only		+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+136	+72
Regional (25 neighbors)		+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+72	N/A

Nomenclature: See separate page